

Dense update at $20\times$



all coordinates receive the step

Mean4 0.340 | severe degradation

Dynamic coordinate support



per-tensor top 40% of adaptive Adam direction

support refreshed every optimization step

Measured at the tested limit

coordinates retained **39.3%**

direction energy retained **85.0%**

support Jaccard **0.600 ± 0.010**

Dense $20\times$ **0.340** $\rightarrow$ **0.445**

Mean4 gain vs dense $20\times$: +0.105